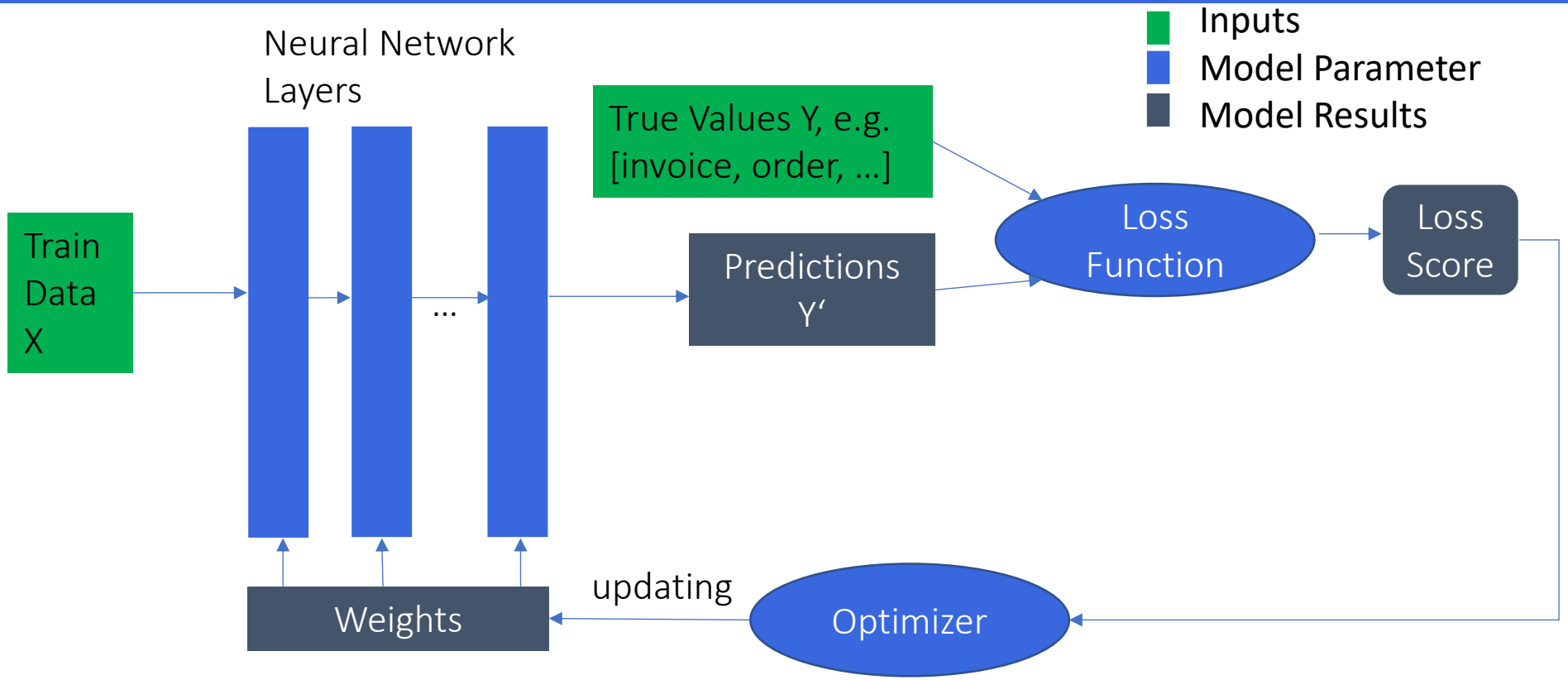
Zero-Shot Classification

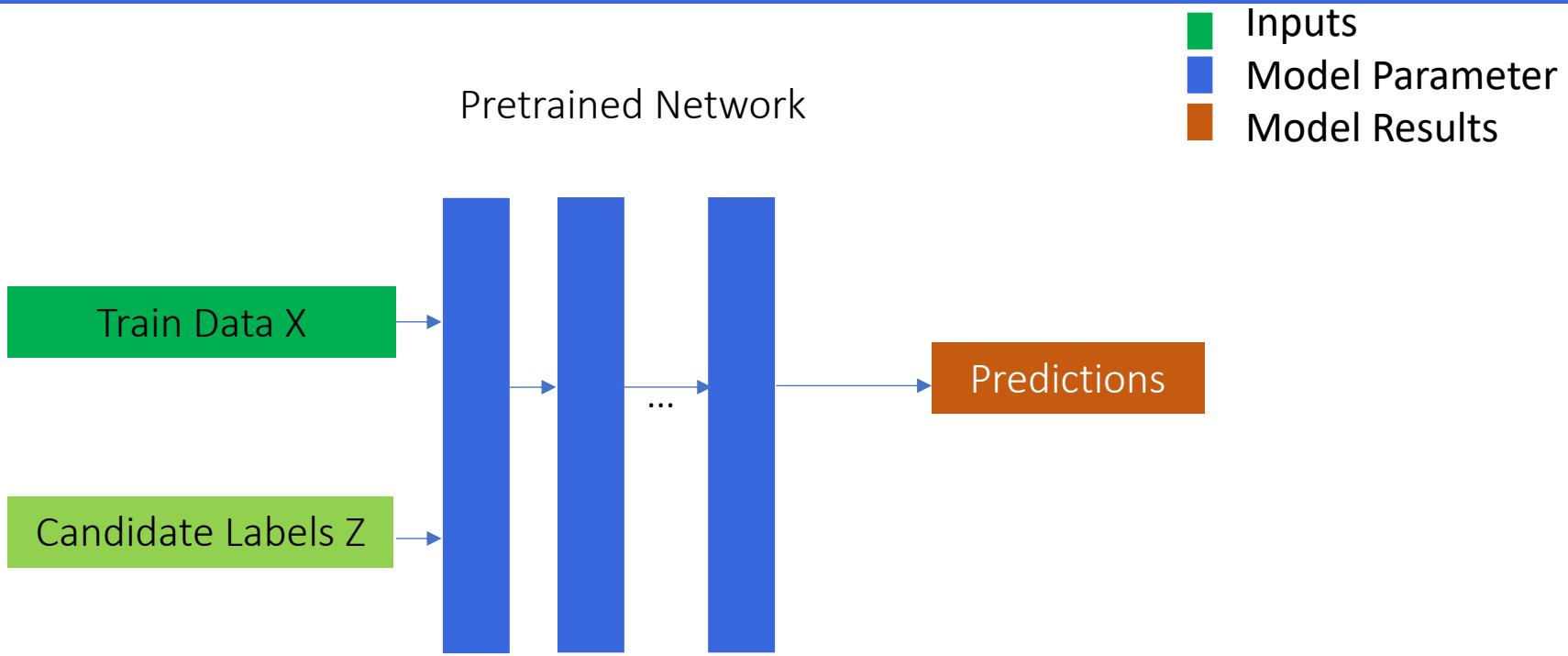
Zero-Shot Classification

Typical Training Process



Zero-Shot Classification

Zero-Shot Text Classification



Zero-Shot Classification

Zero-Shot Goal

Goal

- apply model to task it was not trained on
- generalize trained model to new and unseen classes
- avoid retraining
- use of additional information like embeddings associated to the classes
- additional information used to make predictions for new classes by relating to known classes
- especially useful, if
  - large number of classes is available
  - classes change frequently

Zero-Shot Classification

Introduction

Attribute-Based

- classes associated to attributes that describe its character, e.g. animals
- attributes possible like „has fur“, „can fly“, …
- training: model learns to relate attributes to classes
- inference: able to predict unseen class by combining attributes

Word Embeddings

Transfer Learning

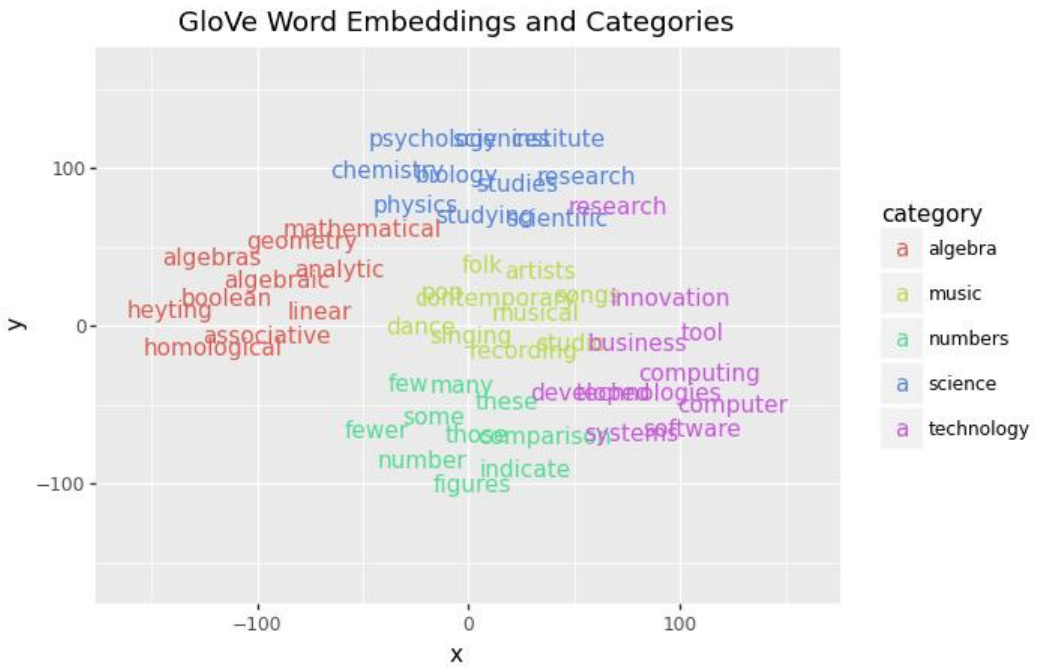
Zero-Shot Classification

Introduction

Attribute-Based

- classes and instances represented in continuous vector space by word embeddings
- embeddings capture semantic relationships between words and concepts
- geometric relationships applied

Word Embeddings



Transfer Learning

Zero-Shot Classification

Attribute-Based

Word Embeddings

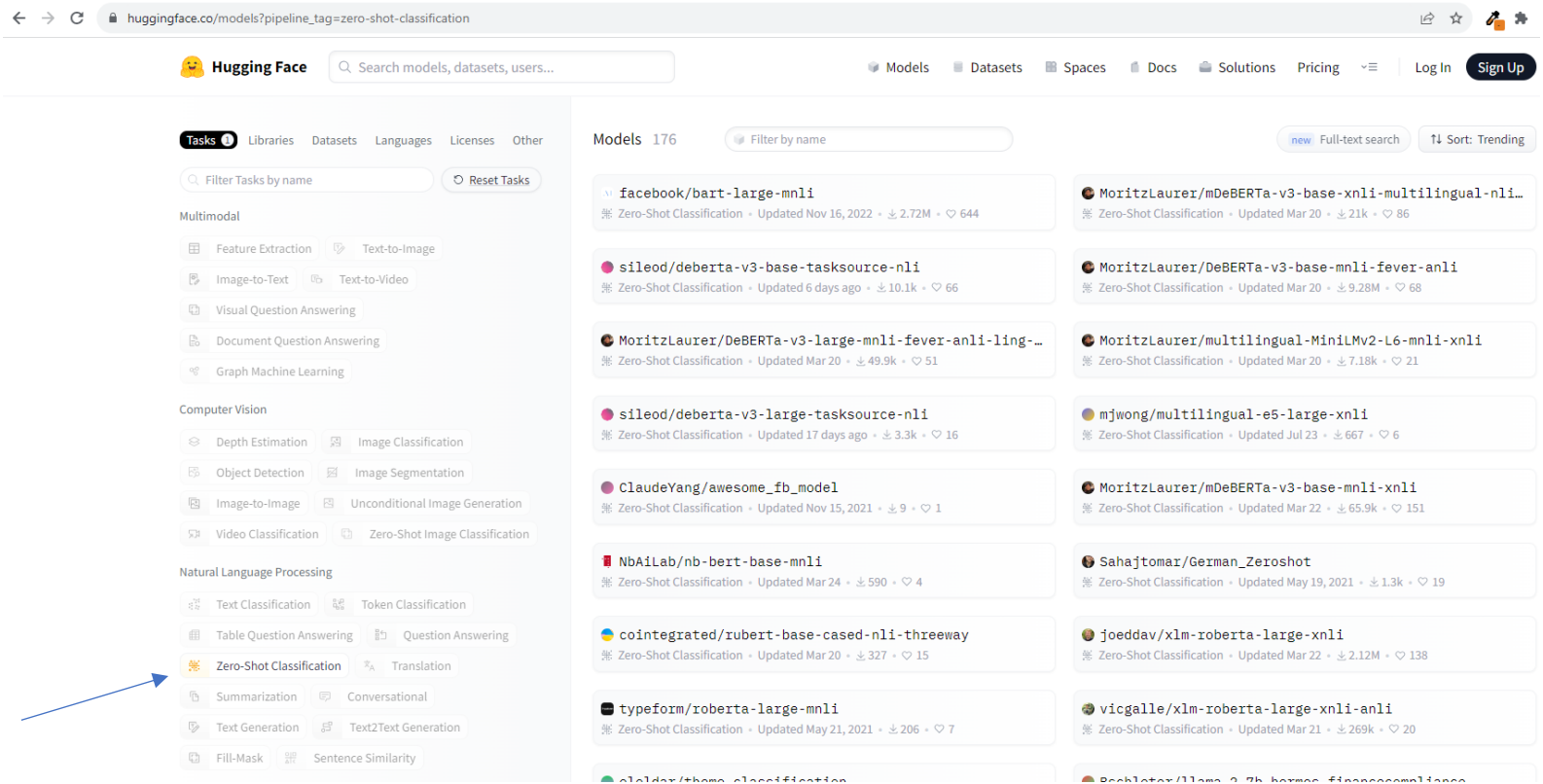
Transfer Learning

- model trained on large dataset
- popular technique „Natural Language Inference“
  - hypothesis tested to be true, false, undetermined given a premise

Zero-Shot Classification

Natural Language Inference

Premise	Candidate Class	Hypothesis	Prediction
"I watched a documentation and it was amazing."	Movie	"The text is about a movie"	0.97
	Book	"The text is about a book"	0.02
	Play	"The text is about a play"	0.01

Zero-Shot Classification

Coding